How the big ideas grew · one row per theme, one column per chapter

Read a row left to right to see how one idea matured across releases (for example pattern matching: instanceof patterns → records → sealed types → switch patterns → primitive patterns).

	1 Java 1.0–1.4	2 Java 5	3 Java 6–7	4 Java 8	5 Java 9–10	6 Java 11	7 Java 12–16	8 Java 17	9 Java 18–20	10 Java 21	11 Java 22–24	12 Java 25	13 Java 26–27
Syntax & ergonomics Less ceremony, same meaning	Exceptions Inner and anonymous classes	Enhanced for loop Autoboxing and unboxing Varargs Static import	Diamond operator Strings in switch try-with-resources Multi-catch and precise rethrow Literals and @SafeVarargs		Local-variable type inference	var in lambda parameters	Switch expressions Text blocks	Always-strict floating point		Unnamed patterns and variables PREVIEW Unnamed classes, instance main PREVIEW String templates ~~WITHDRAWN~~	Unnamed variables and patterns Flexible constructor bodies PREVIEW Compact files & module imports PREVIEW String templates withdrawn ~~WITHDRAWN~~	Compact source files & main() Module import declarations Flexible constructor bodies	
Type system & data modelling Generics, enums, records, sealed	Classes and inheritance Interfaces	Generics Wildcards and PECS Type erasure Covariant return types Enums		Default and static interface methods	Private interface methods		Records Sealed classes PREVIEW	Sealed classes and interfaces ADT: records + sealed + switch		Data-oriented programming			
Pattern matching From instanceof casts to deconstruction							Pattern matching for instanceof	Pattern matching for switch PREVIEW	Record patterns PREVIEW Pattern matching for switch PREVIEW	Record patterns Pattern matching for switch	Primitive types in patterns PREVIEW	Primitive types in patterns PREVIEW	Primitive types in patterns PREVIEW
Functional & streams Lambdas, streams, gatherers				Lambda expressions Functional interfaces Method references Stream API Collectors	Stream and Optional additions	Small API additions	Stream and Collector additions				Stream gatherers		
Collections & APIs Everyday library improvements	Collections framework Iterator, Comparator, utilities Vector, Hashtable, Enumeration ~~DEPRECATED~~ Regular expressions	Scanner, printf, StringBuilder	Deque, ArrayDeque, NavigableMap NIO.2 file system API Small java.util additions Timsort as the default sort	Optional java.time Map helpers Comparator combinators Strings, Base64 and friends	Collection factory methods	HTTP Client New String methods Files.readString and Path.of	String additions Sockets redone, AF_UNIX channels	Enhanced pseudo-random numbers HexFormat		Unicode 15 and small APIs Sequenced collections		Stable Values PREVIEW	HTTP/3 in the HTTP Client Lazy Constants PREVIEW

Tags: preview / incubator features are not final and may change; "final" features carry no tag.

How the big ideas grew · one row per theme, one column per chapter

Read a row left to right to see how one idea matured across releases (for example pattern matching: instanceof patterns → records → sealed types → switch patterns → primitive patterns).

	1 Java 1.0–1.4	2 Java 5	3 Java 6–7	4 Java 8	5 Java 9–10	6 Java 11	7 Java 12–16	8 Java 17	9 Java 18–20	10 Java 21	11 Java 22–24	12 Java 25	13 Java 26–27
Concurrency Threads, executors, virtual threads	Threads and synchronized	ExecutorService and Future Concurrent collections and locks Java Memory Model (JSR 133)	Fork/Join framework	Parallel streams CompletableFuture Concurrency utilities update	Flow API and CompletableFuture VarHandles and StackWalker				Virtual threads PREVIEW Structured concurrency INCUBATOR Scoped values INCUBATOR	Virtual threads Structured concurrency PREVIEW Scoped values PREVIEW	Virtual threads: no more pinning	Scoped values Structured concurrency PREVIEW	Structured concurrency PREVIEW
Runtime, GC & startup JIT, GCs, CDS, AOT	JVM and bytecode Garbage collection HotSpot JIT		invokedynamic G1 garbage collector Java 6 HotSpot performance	Metaspace replaces PermGen	G1 as the default collector Compact strings, indy concat Unified JVM logging AppCDS and container awareness	Nestmates and dynamic constants Low-overhead heap profiling ZGC EXPERIMENTAL Epsilon GC EXPERIMENTAL	Helpful NullPointerExceptions ZGC and Shenandoah production		Reflection via method handles Finalization deprecated ~~DEPRECATED~~	Generational ZGC Vector API, sixth incubator INCUBATOR	AOT class loading & linking Compact object headers EXPERIMENTAL G1, ZGC and Shenandoah updates	AOT cache ergonomics & profiles Compact object headers Generational Shenandoah Vector API INCUBATOR	AOT object caching with any GC G1 throughput: two card tables G1 default in all environments Compact headers by default Vector API INCUBATOR
Platform & tooling Modules, launchers, Javadoc, JFR	Reflection and JavaBeans Serialization NIO buffers and channels assert and java.util.logging	Annotations Writing annotations JMX and java.lang.management	Scripting and Compiler APIs Pluggable annotation processing Console, JDBC 4.0, ServiceLoader	Nashorn JavaScript engine ~~REMOVED~~	Java Platform Module System JShell Multi-release JARs Version scheme and cadence Process API HTTP/2 client (incubator) INCUBATOR	Single-file source launcher Flight Recorder Java EE and CORBA removed ~~REMOVED~~ Nashorn and Pack200 deprecated ~~DEPRECATED~~ Licensing and distributions	Preview features JFR event streaming Strong encapsulation by default jpackage	Strongly encapsulated internals Removals: RMI Activation, jaotc ~~REMOVED~~ Upgrade 8/11 → 17: what breaks LTS every two years macOS: Metal and AArch64 port	Preview and incubator lifecycle Linux/RISC-V port UTF-8 by default Simple web server Code snippets in Javadoc Internet-address resolution SPI	Agent and 32-bit port warnings	Class-File API Launcher, Javadoc and jlink	JFR: CPU-time, sampling, tracing 32-bit x86 port removed ~~REMOVED~~ Upgrading 21 → 25: what you gain	JFR in-process data redaction Prepare to make final mean final Applet API removed ~~REMOVED~~ Reading a release in a week
Native & memory access JNI to the Foreign Function & Memory API							Foreign memory and linker INCUBATOR Vector API INCUBATOR	Foreign Function & Memory API INCUBATOR Vector API (second incubator) INCUBATOR	Foreign Function & Memory API PREVIEW Vector API (3rd–5th incubator) INCUBATOR	FFM API, third preview PREVIEW	Foreign Function & Memory API Unsafe and JNI warnings ~~DEPRECATED~~		
Security & crypto TLS, KEMs, post-quantum						TLS 1.3 and new crypto		Deserialization filters Security Manager deprecated ~~DEPRECATED~~		Key Encapsulation Mechanism API	Post-quantum crypto and KDF API Security Manager, 32-bit retired ~~REMOVED~~	Key Derivation Function API PEM encoding API PREVIEW	Post-quantum hybrid TLS PEM API, previews two and three PREVIEW

Tags: preview / incubator features are not final and may change; "final" features carry no tag.